Code No: 133BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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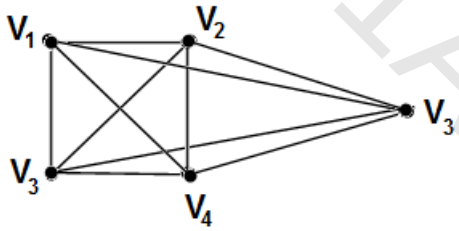
1.a) Use indirect method to show that P → Q , Q → R , ~ (P Λ R), P V R ⇒ R.
b) Obtain principal conjunctive normal form for the formula p V (~ p → (q V (~q → r))) [7+8]

2. Prove that non empty subset H of a group G is a subgroup of G if and only if $ab^{-1} \in H$, for all $a, b \in H$ [15]

3. Let X be the set of all three digit integers i.e $X = \{x \text{ is an integer} / 100 \le x \le 999\}$. If A_i is the set of integers, whose i^{th} digit is i, compute the cardinality of $A_1 \cup A_2 \cup A_3$. [15]

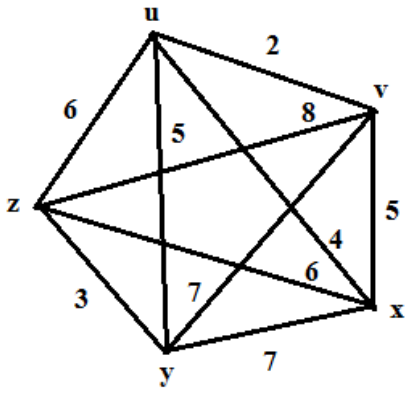
4. Solve the recurrence relation. $u_{n+2} - 2u_{n+1} + u_n = n^2 2^n$. [15]

5.a) Is the following graph planar? If so draw the planar graph 1.



Graph: 1

b) Find the minimum spanning tree shown in Graph 2. [7+8]



Graph: 2

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6.a) Show that the proposition $(p \vee \neg q) \wedge (\neg p \vee \neg q) \vee q$ is a tautology.

b) Prove that the argument is valid.
No mathematicians are fools.
No one who is not a fool is an administrator.
Sita is a mathematician
$\therefore$ Sita is not an administrator. [7+8]

7. The set Q of all rational numbers, the operation is defined by $a*b = a+b-ab$. Show that under this operation Q forms a commutative monoid. [15]

8. Find the coefficient of x^4y^7 in the expansion of $(x-y)^{11}$. [15]

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